



Department of Computer Science and Engineering

Academic Year 2021 – 2022(~~Odd~~ / Even Semester)

Degree, Semester & Branch: B.E, VI & CSE

Course Code & Title: CS8603 & Distributed Systems

Name of the Faculty member (s): Dr.M.Gomathy Nayagam

Innovative Practice Description

- **Unit / Topic:** Unit II / Snapshot algorithms for FIFO channels
- **Course Outcome:** CO 2
- **Topic Learning Outcome:** TLO5
- **Activity Chosen:** Think-Pair Share

- **Justification:**

After teaching the concept of Snapshot algorithms for FIFO channels, this task is given to evaluate students' understanding of the Snapshot algorithm's step-by-step approach.

- **Time Allotted for the Activity:** 20 Minutes
- **Details of the Implementation:**

The course Instructor explained the concept of Snapshot algorithm for FIFO channels. During the explanation, he instructed the students to learn the step by step procedure of Snapshot algorithm with their own examples. Also, he identified pairs of students to think 01 minute about the snapshot algorithm with their own example and they discussed their idea with their pairs about another 01 minute. The pair of students shared their algorithms with their own example in front of the class around 03 to 04 minutes. The following fig 1.a. shows the students in the class think about their own example to explain the snapshot algorithm with their pairs. Fig.1b shows sample pairs of the students come to front and explained the step by step procedure of snapshot algorithm with their own example.



Fig.1a. Students think about the Concept with their peer members



Fig.1b. Two pairs of students (Sample) explained the concept to the class

• **CO – PO / PSO mapping:**

CO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2	PSO 3
CO 2	3	2	1	1	-	-	-	2	-	2	-	2	3	3	2
(1 – Low 2 – Moderate 3 – High)															

• **PO / PSO mapped:**

Innovative practice	PO 1	PO 2	PO 3	PO 4	PO 8
	3	2	1	1	2
Justification for correlation	Students could be able to apply basic engineering theory and principles of Snapshot algorithm	Students could be identify their own example to explain the algorithm	Students could be able to recollect the basic principles of Snapshot algorithm	Students could able to interpret and explain the snapshot algorithm	Students interacted and shared their views about the concept with their peer group by ethically

Innovative practice	PO 10	PO 12	PSO1	PSO 2	PSO 3
	2	2	3	3	2
Justification for correlation	Students could be able to communicate with their	Students could be able to learn the basic engineering concept to	Students could be able to use the snapshot algorithm for implementing	Students could be able to apply Global state in AI	Students could be able to use global state concept in IOT networks.

	peer members and also they shared their views about the concept in front of the class to avoid their stage fear.	understand the working of snapshot algorithm	global state in the mobile computing technologies		
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• **Reflective Critique:**

❖ **Feedback of practice from students and other stakeholders:**

Students felt that this activity is very useful and got good idea about the snapshot algorithm

❖ **Benefit of the practice:** (E.g.: Outcome attainment would have increased due to innovative practice over conventional practice)

- Most of the pairs of students explained the step by step procedure of snapshot algorithm in correct way of their own example.
- Students interaction with their peer members and the class during the activity was good and effective.

❖ **Challenges faced in implementation:**

- Due to time limitation, only few pairs of students were made to do the presentation in front of the class by random call.

References:

- ❖ <https://www.kent.edu/ctl/think-pair-share>
- ❖ Johnson, D., Johnson, R. (1999). Making Cooperative Learning Work: Theory into Practice, 38 (2), 67-73. Fitzgerald, D. (2013). Employing think-pair-share in associate degree nursing curriculum. Teaching & Learning in Nursing. 8, 3, p 88-90.
- ❖ Goodwin, M. (1999). Cooperative learning and social skills: What skills to teach and how to teach them. Intervention in School & Clinic, 35 (1), 29.
- ❖ Lyman, F. (1981). The responsive classroom discussions: the inclusion of all students. A. Anderson (Ed.), Mainstreaming Digest, College Park: University of Maryland Press, pp. 109-113.
- ❖ Mr.R.Venkatesh, "Introduction to Yoga and Meditation for professional excellence and stress management " in course GE 6075 Professional Ethics in Engineering, <https://www.ritrjpm.ac.in/images/computer-science/TPS-Venkat-GE6075.pdf>

Signature of Faculty Member

HOD